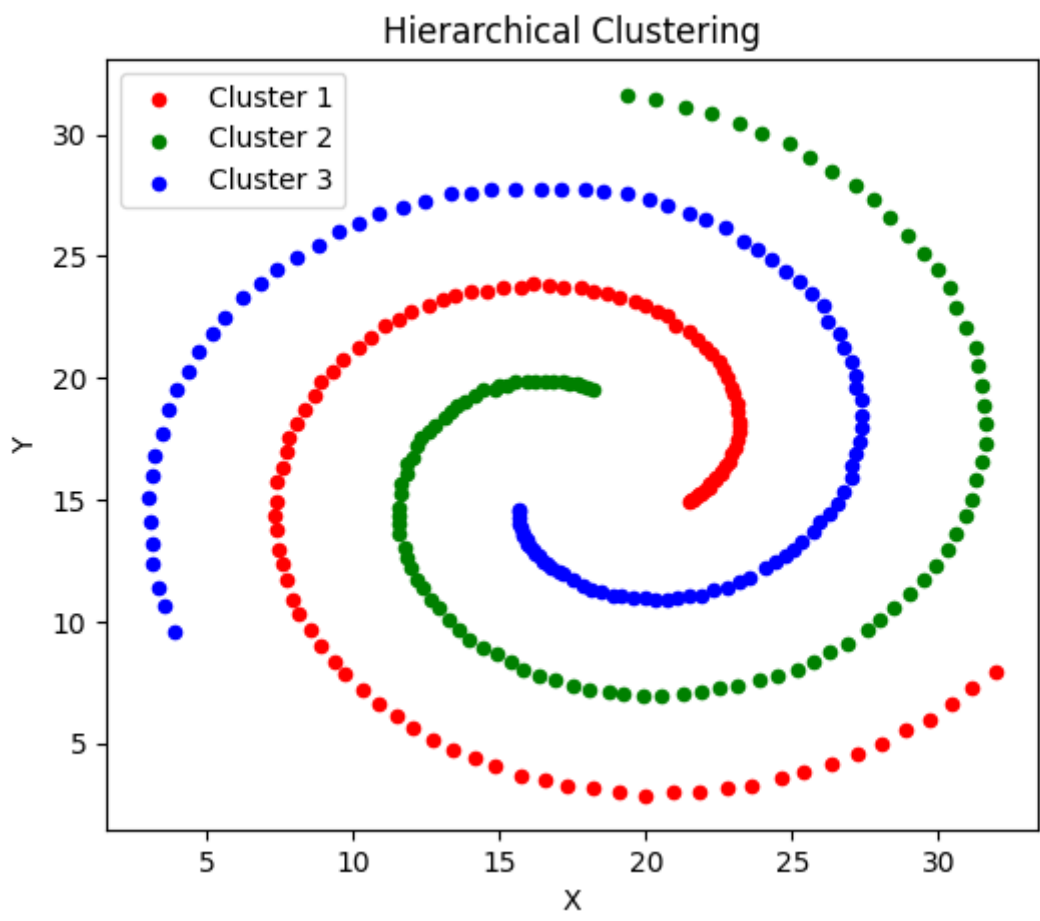
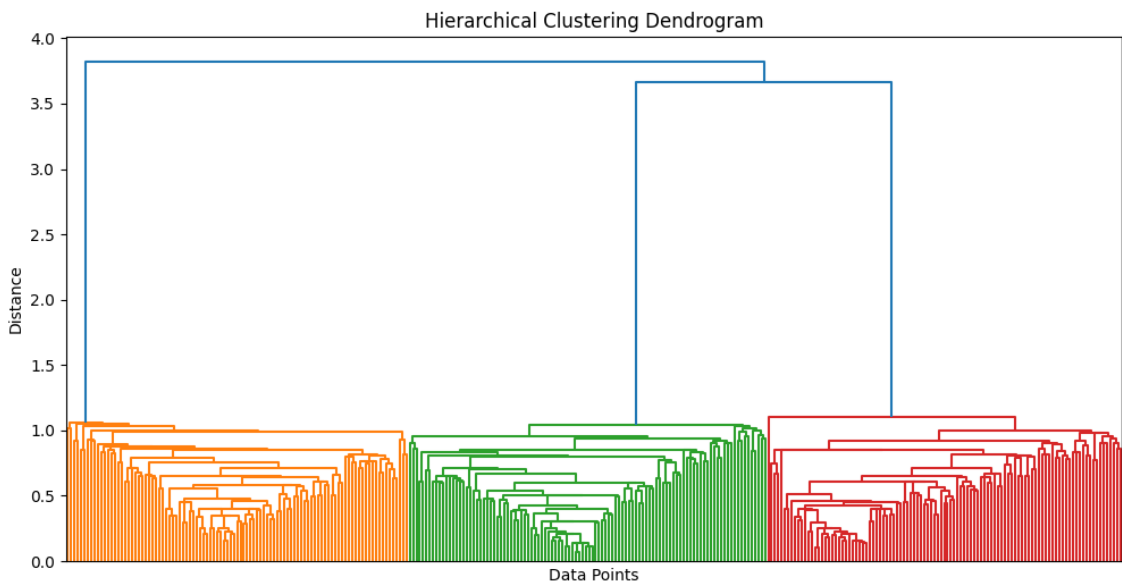


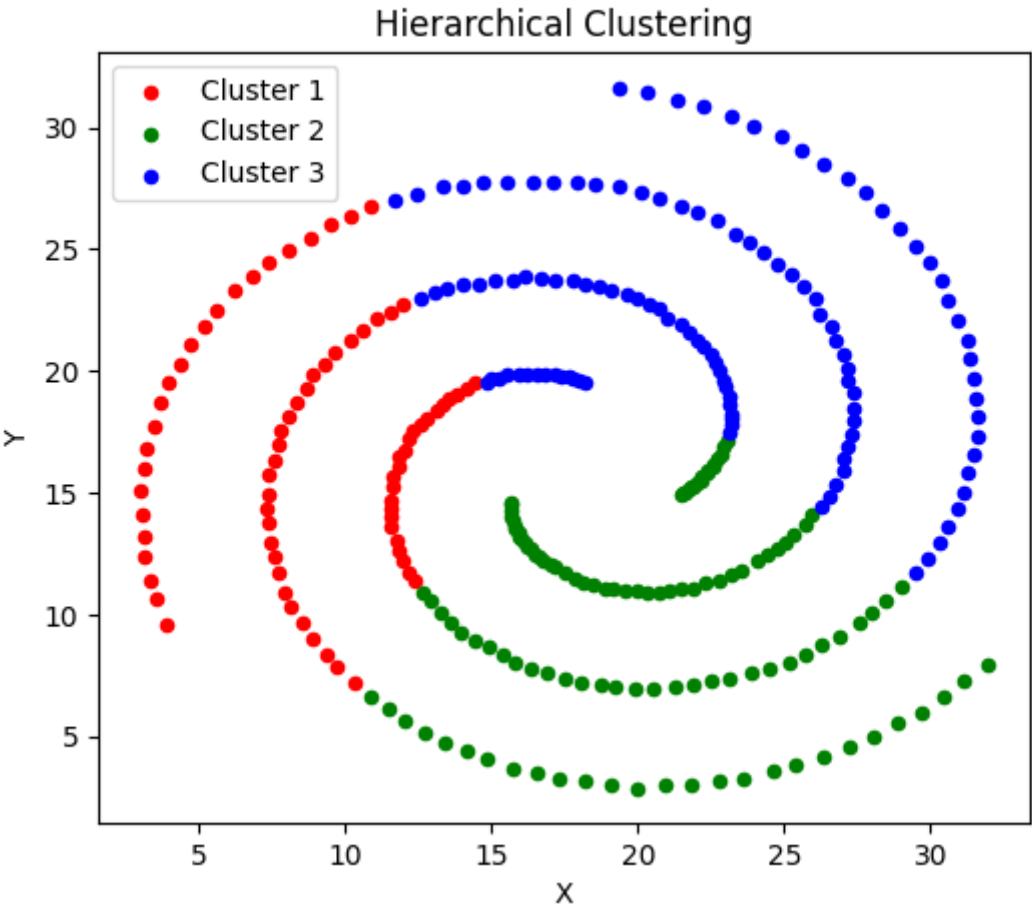
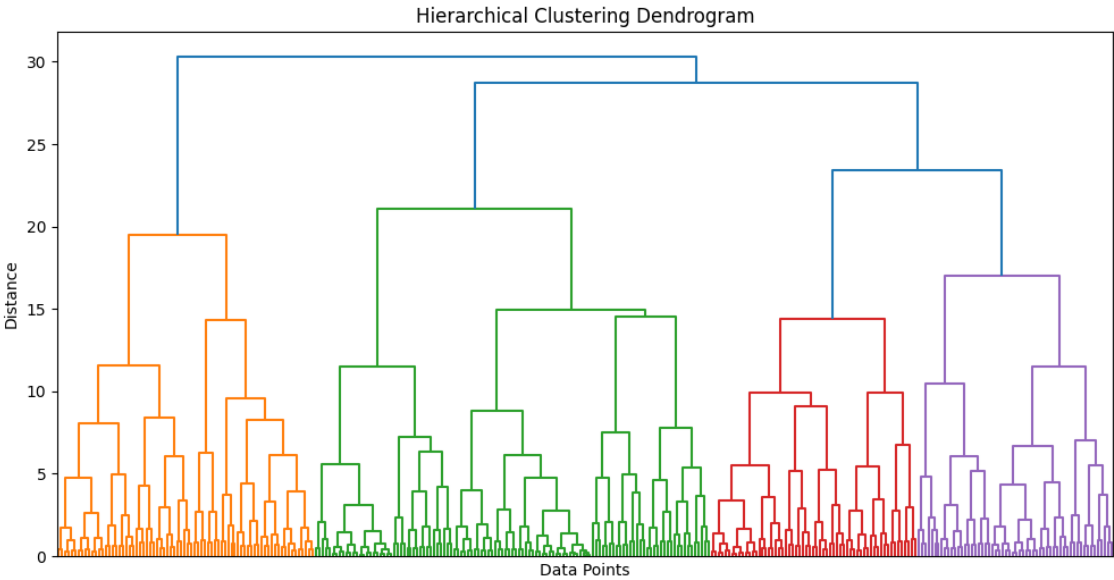
Single Linkage Clustering

SSE: 30109.3504
Rand Index: 1.0000
Cophenetic Correlation Coefficient: -0.0502
Silhouette Score: 0.0013



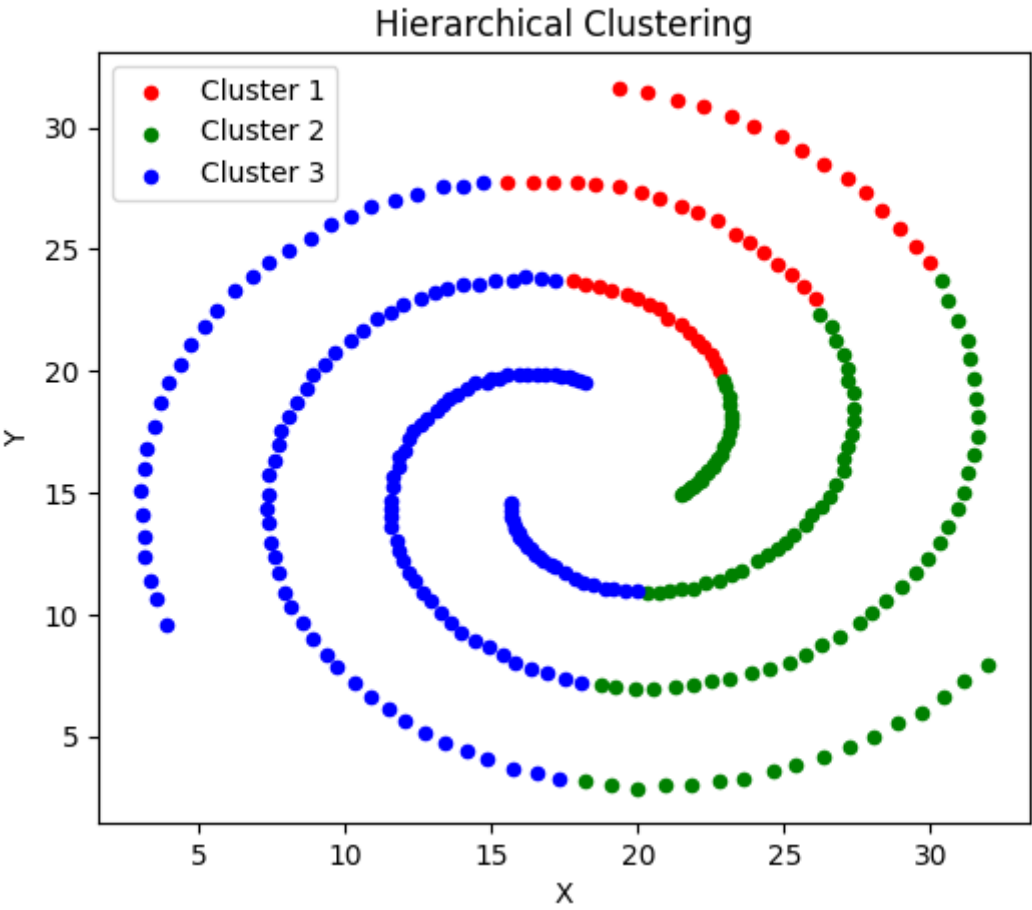
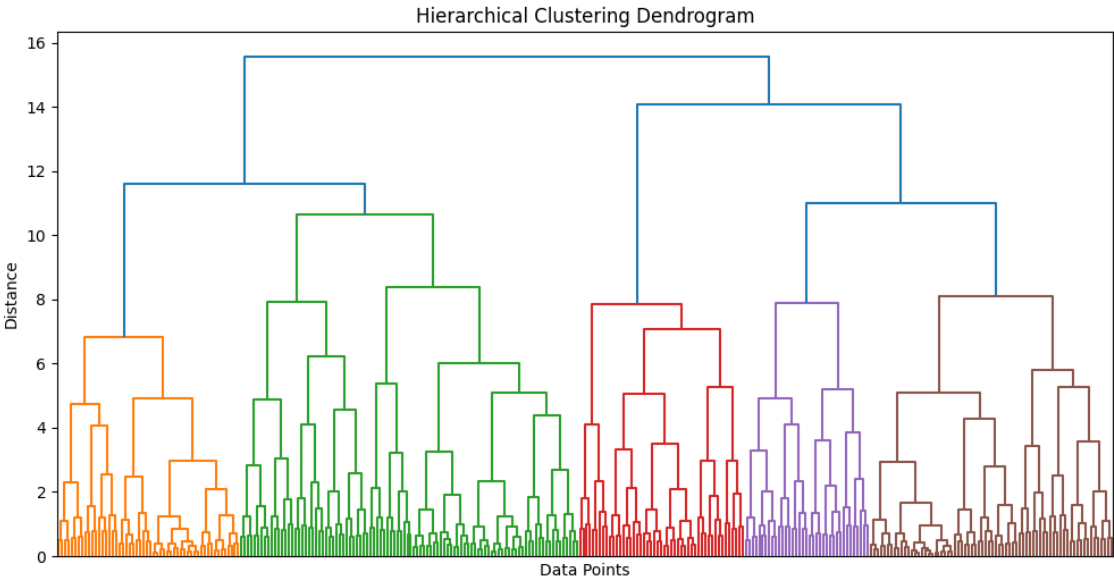
Complete Linkage Clustering

SSE: 13004.3742
Rand Index: 0.5537
Cophenetic Correlation Coefficient: -0.1495
Silhouette Score: 0.3455



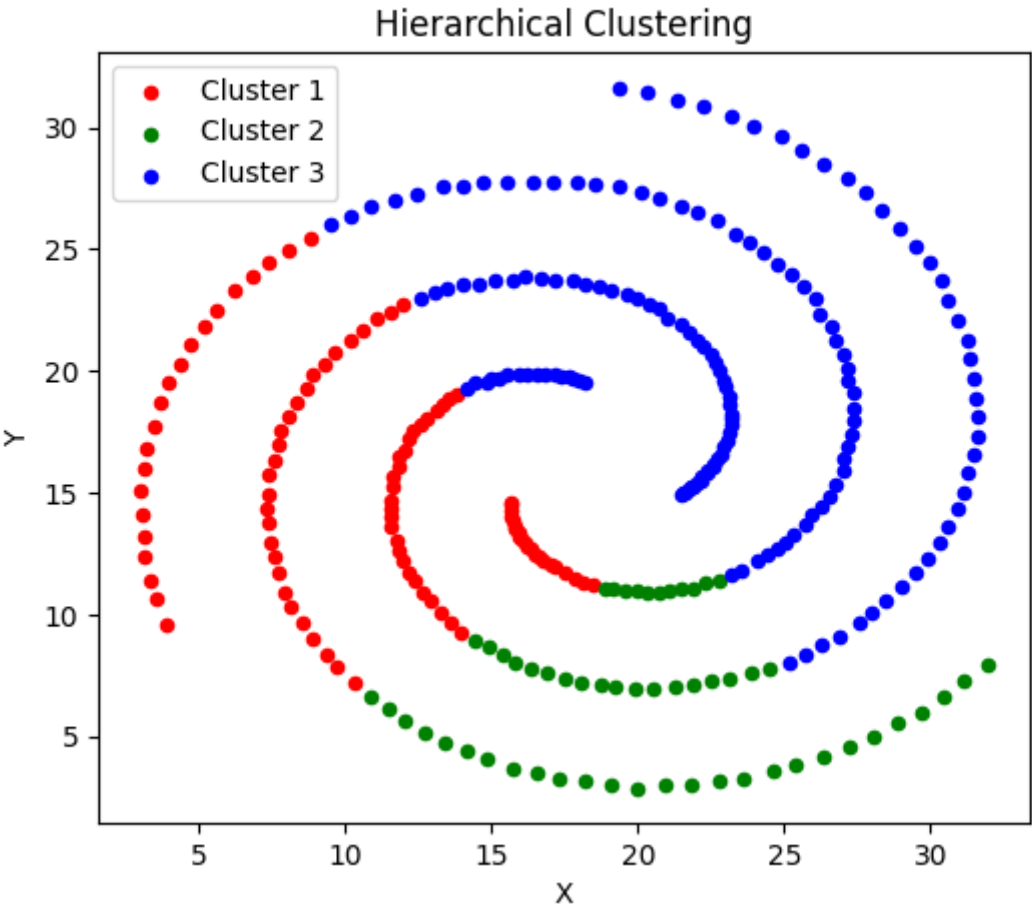
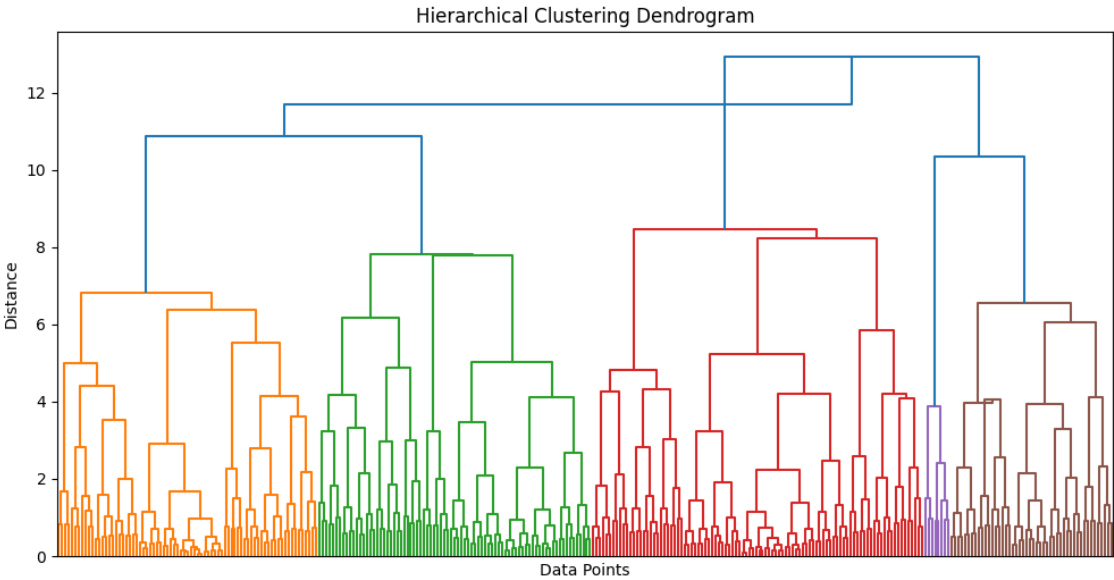
Average Linkage Clustering

SSE: 14132.1538
Rand Index: 0.5366
Cophenetic Correlation Coefficient: -0.3220
Silhouette Score: 0.3185



Centroid Linkage Clustering

SSE: 14636.1893
Rand Index: 0.5422
Cophenetic Correlation Coefficient: -0.0717
Silhouette Score: 0.3055



Analysis of Results

Quick note:

- The dendrograms for linkage methods (complete, average, centroid) have more than 3 colors, but these do not equate to clusters. The SciPy dendrogram function assigns the colors based on a distance threshold. Looking at the actual plot of the clustering, we can see that there are only 3 clusters.

Best SSE: Complete Linkage

Best RI: Single Linkage

Best CCC: Single Linkage

Best Silhouette Score: Complete Linkage

- Looking at the metrics, complete or single could be the best linkage method. However, for the spiral dataset, the only useful metric is the Rand Index, which shows the similarity between our clustering and the ground truth. The other metrics are misleading due to the spiral nature of the clusters.
- Single Linkage had a perfect score on Rand Index (1), which means that our algorithm assigned every point to the correct cluster. This can also be seen by viewing the plot of the spiral.
- While the other linkage methods may have performed better on certain metrics. By looking at the plots we can see that clustering is not accurate.